

Western Blot Integrity Atlas

Full reference-proteome molecular-weight audit report with embedded figures and complete tables

Defensive use only. This report helps evaluate whether a claimed western blot result is plausible and what controls are needed. It must not be used to plan, enable, or disguise blot substitution or fabrication.

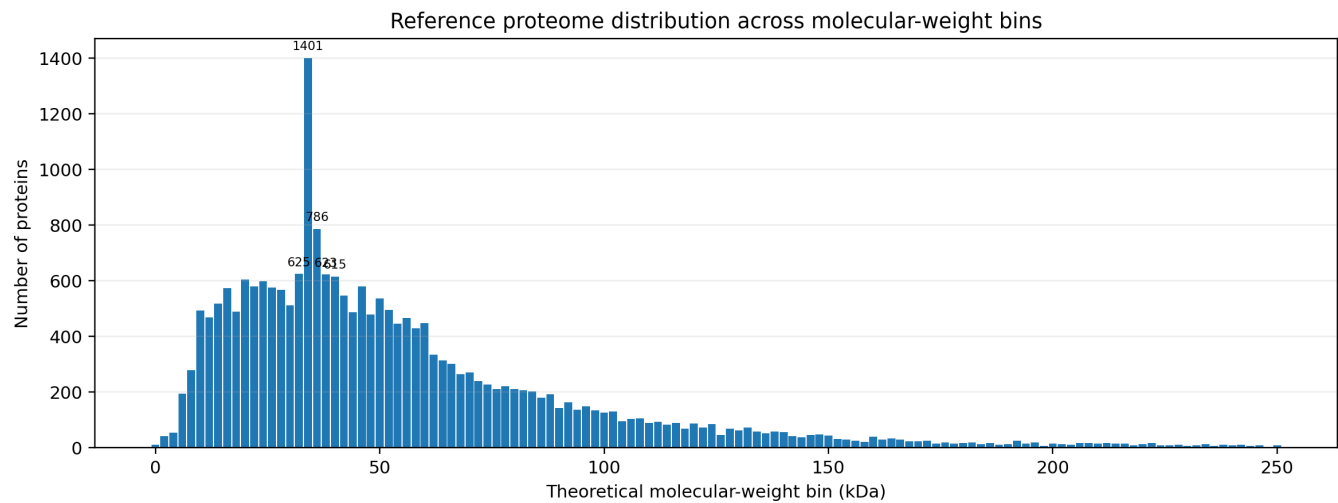
Metric	Value
Proteins parsed	22,714
Molecular-weight bins	250
Bin width	2 kDa
MW range	0.97 to 862.79 kDa
Median MW	44.81 kDa
Mean MW	59.95 kDa
Proteins <= 50 kDa	12,704 (55.9%)
Proteins 50-100 kDa	6,928 (30.5%)
Proteins > 100 kDa	3,082 (13.6%)
High-priority bins	25
Elevated-priority bins	38
Routine-priority bins	187
Source FASTA	cow_UP000009136_9913.fasta

Method in one breath

For each protein sequence, the script computes theoretical average molecular weight from residue masses, assigns a molecular-weight bin, applies the external marker, signaling-prefix, and family-pattern annotations, maps supplied public malpractice cases to bins, and calculates an audit-burden priority. Similar apparent molecular weight is never treated as proof of identity; it only increases the need for full-blot provenance and orthogonal controls.

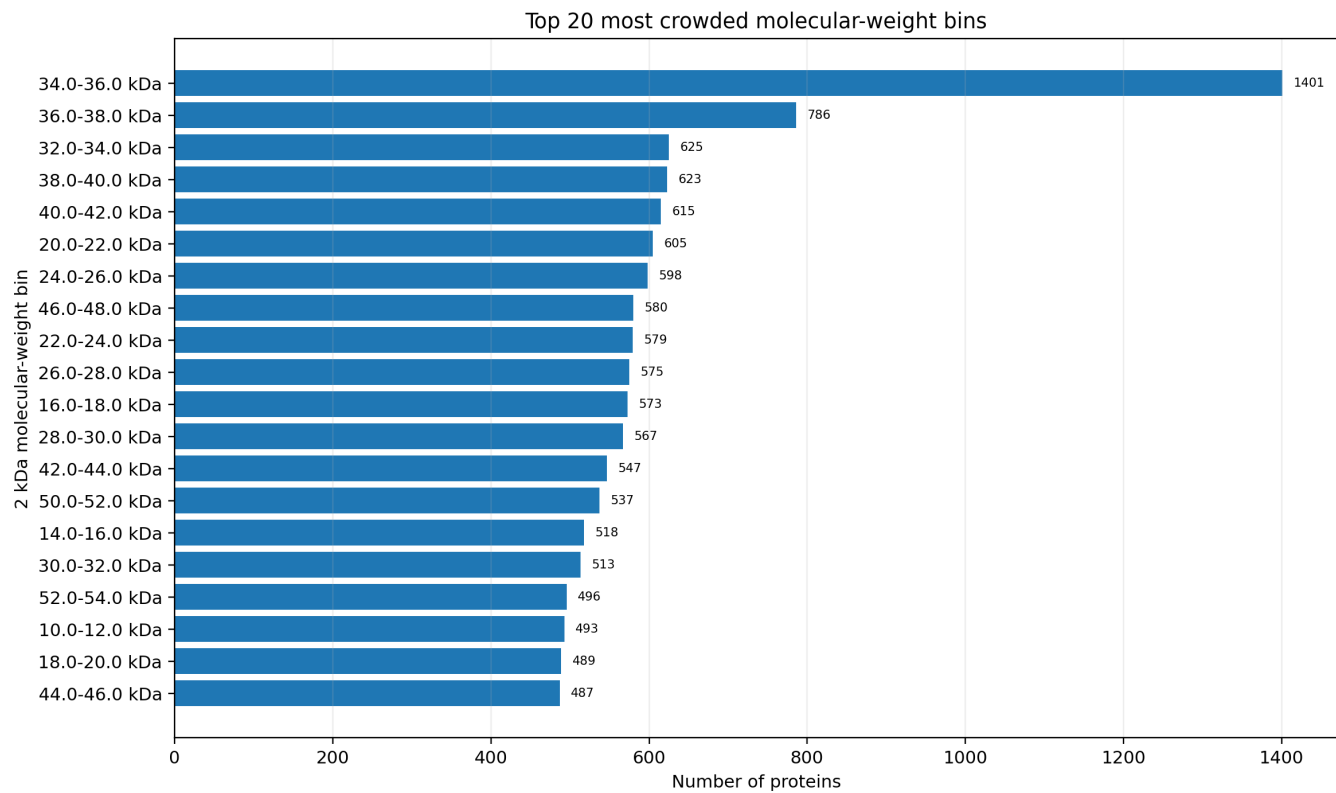
Molecular-weight distribution of the reference proteome.

Proteins were grouped into theoretical molecular-weight bins. Dense gel regions are especially relevant for western blot review because many unrelated proteins occupy nearby molecular-weight neighborhoods.



Most crowded molecular-weight neighborhoods.

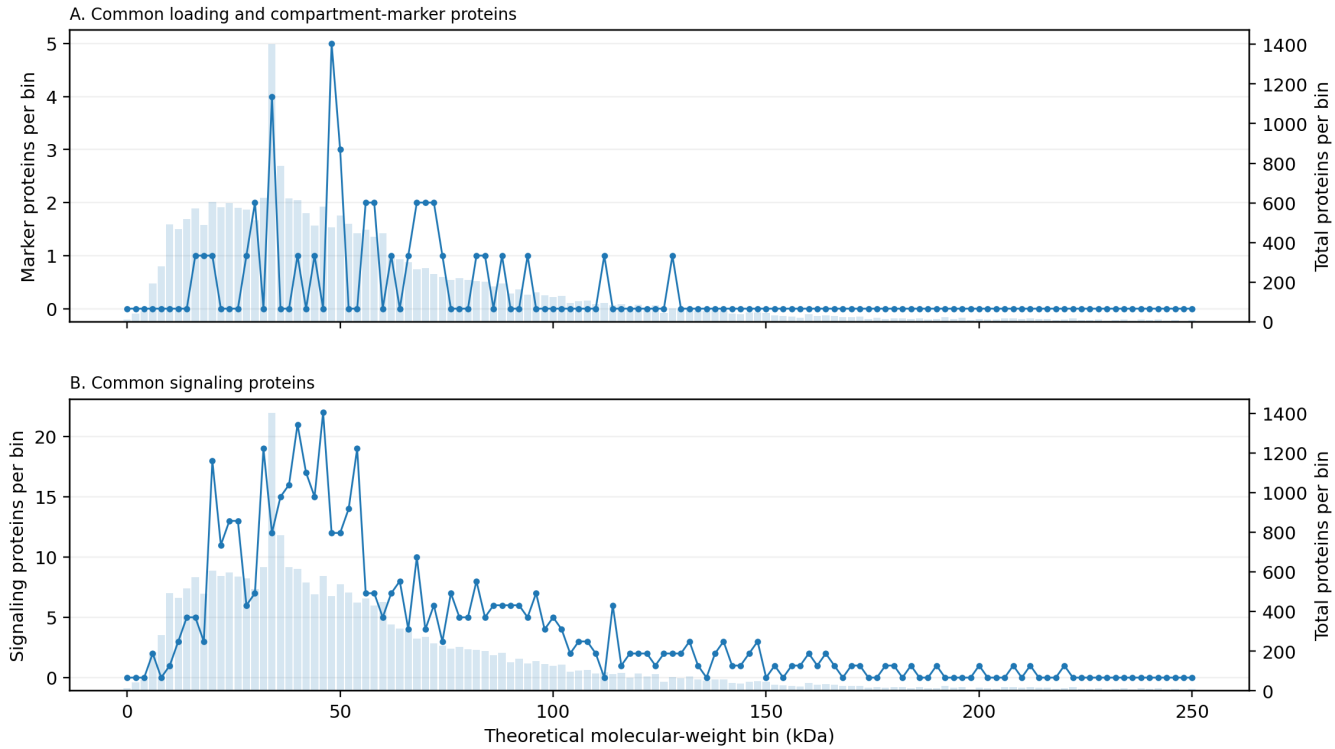
The top bins are ranked by protein count. These bins are not suspicious by themselves; they indicate where molecular weight alone is a weak identity criterion and where validation documentation becomes more important.



Distribution of common western blot markers and signaling proteins across molecular-weight bins.

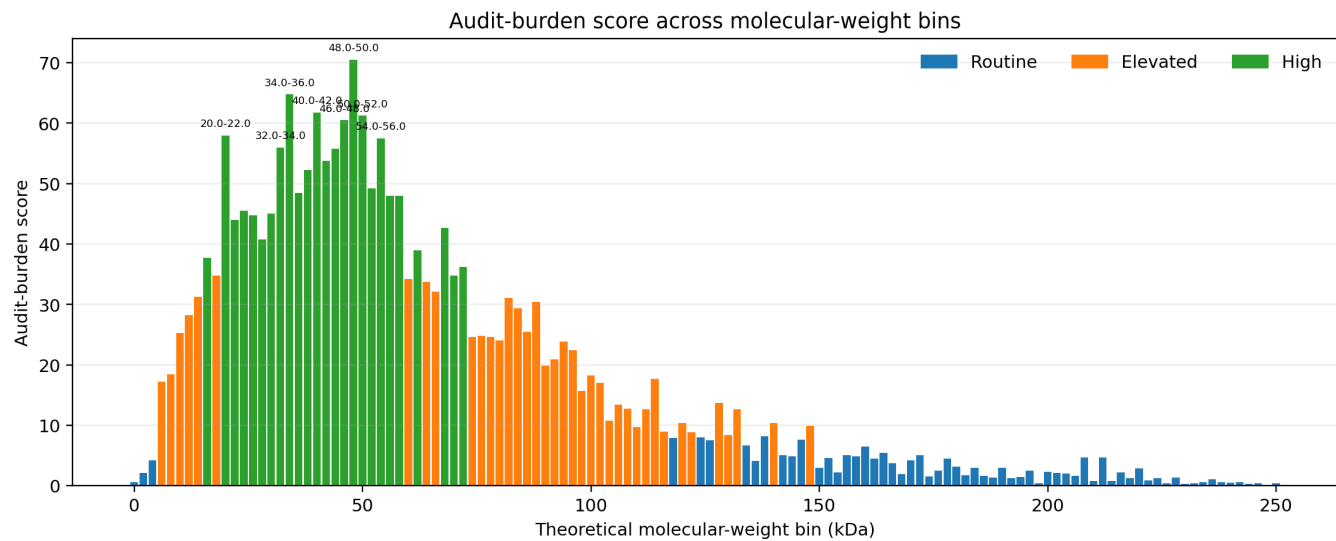
The pale background bars show total protein density, while the overlaid lines show audit-relevant annotations. Panel A shows configured loading or compartment markers; Panel B shows configured signaling-prefix matches.

Audit-relevant annotations across molecular-weight bins



Audit-burden score across molecular-weight bins.

The score integrates protein density, marker count, signaling-prefix count, and configured family/context information. High-scoring bins define high-documentation zones for western blot identity claims.



Highest-audit molecular-weight neighborhoods: score summary

MW bin	N	Markers	Signal	Priority	Score
48.0-50.0 kDa	478	5	12	high	70.5
34.0-36.0 kDa	1401	4	12	high	64.8
40.0-42.0 kDa	615	1	21	high	61.8
50.0-52.0 kDa	537	3	12	high	61.2
46.0-48.0 kDa	580	0	22	high	60.5
20.0-22.0 kDa	605	1	18	high	58.0
54.0-56.0 kDa	446	0	19	high	57.5
32.0-34.0 kDa	625	0	19	high	56.0
44.0-46.0 kDa	487	1	15	high	55.8
42.0-44.0 kDa	547	0	17	high	53.8
38.0-40.0 kDa	623	0	16	high	52.2
52.0-54.0 kDa	496	0	14	high	49.2
36.0-38.0 kDa	786	0	15	high	48.5
56.0-58.0 kDa	466	2	7	high	48.0
58.0-60.0 kDa	429	2	7	high	48.0
24.0-26.0 kDa	598	0	13	high	45.5
30.0-32.0 kDa	513	2	7	high	45.0
26.0-28.0 kDa	575	0	13	high	44.8
22.0-24.0 kDa	579	0	11	high	44.0
68.0-70.0 kDa	264	2	10	high	42.7
28.0-30.0 kDa	567	1	6	high	40.8
62.0-64.0 kDa	334	1	7	high	39.0
16.0-18.0 kDa	573	1	5	high	37.8
72.0-74.0 kDa	240	2	6	high	36.2
70.0-72.0 kDa	271	2	4	high	34.8

Highest-audit molecular-weight neighborhoods

These bins combine protein density with configured markers, signaling-prefix matches, and family-rich regions. They are not substitution recommendations; they are places where reviewers should be most insistent about source files and controls.

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
48.0-50.0 kDa	478	5	12	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	high	PICK1, ECMX, PSKH1, IFRD2, SYT11, TEDC2 [Tubulin family], STK25, SERPINB12, CALR [ER markers], BZW1, TES, MRPL37, TRPC2, LOC100299281, FOXD2, , OAT [Mitochondrial markers], [ER markers]
34.0-36.0 kDa	1401	4	12	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	LOC526989, MS4A1, IGFBP2, VTA1, OR5B32, DHRS1, GPM6B, CCDC69, CDK1 [Cell-cycle CDK/cyclin], TMX2, TAL1, TMEM255B, GLRX3, OR4K36, LOC530031, LOC100850276, PRSS27, OR8510
40.0-42.0 kDa	615	1	21	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	high	CD44, , PGA, SLC35C1, ACKR4, ST3GAL2, ZDHHC12, TIMM50 [Mitochondrial markers], GMPR2, ERI1, HRH2, RBM4B, BCL2A1 [Apoptosis caspase/BCL], MRPS25, CHAC2, , MXRA7, ARF6, ASCL5, FAM219B, EMC4, LOC783993
50.0-52.0 kDa	537	3	12	Actin family; Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Nuclear receptor PPAR; Tubulin family	high	TMEM184C, MEIKIN, RCC1L, C20H5orf22, ZNF362, OR629, CHGA, SOX10, NDE1, GSDMA, MOCS3, ERFF1, TUBA8 [Tubulin family], LOC782966 [Tubulin family], MSR1, HSF4 [Heat shock proteins], PRUNE1, TEX13B
46.0-48.0 kDa	580	0	22	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family	high	, POGUT1, BEND7, SERPINA7, RPL3, NCEH1, GNA13, SERPINI2, FOXF2, NR13, HRH4, DPAGT1, SSTR3, HABP4, HEPACAM, NFATC2IP [Actin family], FCRLB, PLA2G15
20.0-22.0 kDa	605	1	18	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; p53 pathway	high	LOC100847317, CDKN2AIPNL [Actin family; Cell-cycle CDK/cyclin], MORN5, IL7, LOC104970886, CATHL2, PARK7, CRYAB, BCL2A1 [Apoptosis caspase/BCL], MRPS25, CHAC2, , MXRA7, ARF6, ASCL5, FAM219B, EMC4, LOC783993
54.0-56.0 kDa	446	0	19	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	WDR37, SCARB2, ANXA11, CASP16 [Apoptosis caspase/BCL], SLC2A3, SLC38A1, WASF2 [Actin family], TRIM21, CPVL, SLC7A5, , NR6A1, CHST7, GALK2, AEBP2, CWC27, MMP13, MAPKAPK5 [MAPK cascade]
32.0-34.0 kDa	625	0	19	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	BOLA-DQB, ZNF691, AGPAT1, NKX2-6, PGAM5 [Mitochondrial markers], DTWD2, SLA2, SAMD10, ASGR2, PYCR3, PNP, GSX2, RPH3AL, TPRG1, ASRGL1, LIX1, SSR1, FOSB
44.0-46.0 kDa	487	1	15	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	PHLDA1, NIPAL4, NCK2, BSCL2, , WTAP, MEI4, PCBP3, MLC1, , CRTAM, PAK1IP1 [Actin family; p53 pathway], SH3GLB2, BABAM1, , PSMC6, WWTR1, LAMP1
42.0-44.0 kDa	547	0	17	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	RPUSD4 [Mitochondrial markers], ELF3, PEX26, ACTA2 [Actin family], ZAR1, LOC112443864, ACTC1 [Actin family], OGG1, TSKU, PTX3, GPR85, GIMAP4, GLUL, VGLL4, LIAS [Mitochondrial markers], LOC515736, PAG12, FUT9
38.0-40.0 kDa	623	0	16	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	NTM, LCMT1, REM2, TMEM268, OTULIN, YOD1, GMPR2, PAQR5, MAGT1, ST6GALNAC6, ELMOD1, LOC789863, KCNK1, NUP35 [p53 pathway], DHRSX, DMRT1, VPS37C, DEGS1
52.0-54.0 kDa	496	0	14	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	high	TIMM44 [Mitochondrial markers], ATG4A, RMND1, LOC786377, ACP1, PKNOX2, MXRA8, WNT2B, ESRRG, SAMM50, F9, INPP5K, GPAT4, CIR1 [Actin family], ZBTB8A, TACR3, LOC520034, GPAT4
36.0-38.0 kDa	786	0	15	AKT pathway; Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; MAPK cascade; Mitochondrial markers; Tubulin family	high	PLCXD1, TMEM231, HNRNPAB, HNRNPA2B1, OR6C7G, STBD1, ASB7, FOXR2, AUH, C19H17orf58, LDHC, SIRPB2, GAS1, ANKRD45, SCTR1, OR10AL43, OR52E2, HS3ST1
56.0-58.0 kDa	466	2	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Heat shock proteins; Histone family; Mitochondrial markers; Nuclear receptor PPAR; Tubulin family; p53 pathway	high	CHRNA7, KCNF1, SLC14A4, USP14, MBOAT1, TUBB3 [Tubulin family], CCDC50, GLRB, PSMO5, PTDSS2, DRC4, CYP27B1, CYP2C282, TENT2, TP42, MRPS34 [Mitochondrial markers], SLC2A2, CYP21 [p53 pathway]
58.0-60.0 kDa	429	2	7	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; ER markers; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	NEPN, GHDC, LOC522334, ANGPTL1, SLC16A3, LOC132344484, ETV3, KCNA10, GLCCI1, MEPE, KRT10, GPT2, CCT6B, ZNF703, KRT4, PAK2, RAB11FIP2 [Actin family], PSAP
24.0-26.0 kDa	598	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	INSL6, NME5, , LOC781977, PDGFA, DPT, LHFLP3, LOC617402, MRPL48, CDK2AP2 [Cell-cycle CDK/cyclin], , DECR2, C4BPB, HMG82, HESX1, AICDA, TNNI3, AICDA
30.0-32.0 kDa	513	2	7	Actin family; Apoptosis caspase/BCL; Histone family; Lamin family; Mitochondrial markers; Tubulin family	high	, PSMB7, , TATDN3, LENG1, PLP1, PRSS33, RPL7L1, RPL7A, ASB13, NSA2, PSMB7, MRPL28, SMUG1, NKX2-2, IMPA1, CD209, BPGM
26.0-28.0 kDa	575	0	13	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	high	UBTD1, C8H9orf152, CCDC124, MRPS18A, ELSBPB1, KCTD11, CSN1S2, MT-CO2 [Mitochondrial markers], , RND1, KCTD11, MOB4, SHISAL1, CDKN1C [Cell-cycle CDK/cyclin], CLDN11, PGRMC2, SSPN, MPDU1
22.0-24.0 kDa	579	0	11	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; Tubulin family; p53 pathway	high	LOC100298530, RHOC, LOC132344423, CLDN11, PEMT [Lamin family], PPIF [Mitochondrial markers], CHMP1B, C17H5orf52, RAB1B, NRASE11, MGMT, STUM, AP3S2, , FGF4, CD7, , DUSP21 [p53 pathway]
68.0-70.0 kDa	264	2	10	Actin family; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel	high	PLK1, BCAM, MCOLN2, C10H14orf39, KNG1, MGAT4B, OTOP1, ARHGEF25, RNF180, FAM200C, LONRF1, ZNF668, USP2, TESK1, HACL2, MINDY2, BIRC2, LOC132342087
28.0-30.0 kDa	567	1	6	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; NF-kB/Rel; Tubulin family; p53 pathway	high	LRRC3, SPINT2, CMBL, LOC505326, , LRRRC61, PDCL2, FAM3B, YIPF5, RPL8, MOG, GMPPB, CLDN18, LOC782057, PIP4P2, N6AMT1, GSC, MRPS26
62.0-64.0 kDa	334	1	7	Actin family; Cell-cycle CDK/cyclin; ER markers; Histone family; Lamin family; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	high	TDRKH, LOC786974, CFAP53 [p53 pathway], LMBRD1 [Lamin family], CYP2U1, XRRA1, EPHX2, SLC1A2, EPN2, USH1C, GNL3, CCM2L, LOC790009, OTOP2, SELENON, PHF1, NOC4L, USH1C
16.0-18.0 kDa	573	1	5	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers	high	PLA2G2E, PLA2G1B, COTL1, JPT1, , C1D, LOC100296105, LAMTOR3, , , MRPL27, , SEC61B, CNIH4, TMEM18, RPS19, HBE1
72.0-74.0 kDa	240	2	6	Actin family; Apoptosis caspase/BCL; ER markers; Heat shock proteins; Histone family; JAK-STAT; Mitochondrial markers	high	DDX43, KIFBP, FUBP1, ODAD3, RAB11FIP4 [Actin family], ZCWPW1, CREBRF, ITK, BCL6 [Apoptosis caspase/BCL], RHOT1 [Mitochondrial markers], LOC516736, COQ8A [Mitochondrial markers], FBXO21, SLC8B1 [Mitochondrial markers], FEM1A, PRDM14, LOC100124497, PRODH
70.0-72.0 kDa	271	2	4	Actin family; Cell-cycle CDK/cyclin; FOXO/SMAD transcription; Heat shock proteins; Lamin family; MAPK cascade; Mitochondrial markers	high	ENG, EYA4, MTM1, RLIM, KLHL31, FNBP1L, MED26, SLC6A4, TKTL2, ACOT11, DMWD, SAMD4A, RMI1, PRICKLE3, KLHL17, SLC6A11, CKAP2, VPS33A
18.0-20.0 kDa	489	1	3	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family	elevated	NCBP2, RFESD, LRCOL1, ALOX5AP, IER3, IQCF5, ACP1, SCOC, LOC505033, MRPL50, GABARAPL2, UBD, KCNE4, UXT, LOC101903026, NFRSF12A, LY6K, SPACA3
60.0-62.0 kDa	448	0	5	AKT pathway; Actin family; FOXO/SMAD transcription; Heat shock proteins; Histone family; Lamin family; Mitochondrial markers; Tubulin family; p53 pathway	elevated	CDC25B, SLC22A7, , GAS8, HNRNPLL, PUF60, MARCOL, CPN2, MARCHF1, CLK2, RBM46, SNRNP27, TRIM55, CREB3L4, CHRNB2, SLC22A11, GBA1, STEAP3
64.0-66.0 kDa	315	0	8	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	PRKAA1, FAM151A, TGFBR2, ARIH1, PINK1, LMF1, LRRIQ4, RBM47, , NXPE3, AATF, TTC24, , LACTBL1, PRPF38B, HNRNPL, TLE6, CCNK [Cell-cycle CDK/cyclin]
66.0-68.0 kDa	302	1	4	Actin family; Cell-cycle CDK/cyclin; ER markers; FOXO/SMAD transcription; Histone family; Lamin family; MAPK cascade; Mitochondrial markers	elevated	, ZNF181, CENPB, RTKN, ASMTL, CCDC38, JPH4, GPR153, POMGNT2, DCAF8, CAMK1G, ZNF143, ACSM5, TOX4, ENC1, F5IP1 [Actin family], CHRM3, PCSK2
14.0-16.0 kDa	518	0	5	Actin family; Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers; p53 pathway	elevated	, TMEM262, , , , H2A3 [Histone family], , CDKN2B [Cell-cycle CDK/cyclin], LOC132343308, , GABARAPL1, , , BATF, LOC514876, LLCFC1

Common loading/compartment markers present in this proteome file

MW kDa	Gene	Accession	Entry	Protein	MW bin
16.30	TOMM20	A6H7B1	TOM20_BOVIN	Mitochondrial import receptor subunit TOM20 homolog	16.0-18.0 kDa
19.57	COX4I1	P00423	COX41_BOVIN	Cytochrome c oxidase subunit 4 isoform 1, mitochondrial	18.0-20.0 kDa
20.24	H2AC1	A0AAA9THX7	A0AAA9THX7_BOVIN	Histone H2A	20.0-22.0 kDa
28.75	PCNA	Q3ZBW4	PCNA_BOVIN	DNA sliding clamp PCNA	28.0-30.0 kDa
30.74	VDAC1	P45879	VDAC1_BOVIN	Non-selective voltage-gated ion channel VDAC1	30.0-32.0 kDa
31.62	VDAC2	P68002	VDAC2_BOVIN	Non-selective voltage-gated ion channel VDAC2	30.0-32.0 kDa
34.37	RPLP0	Q95140	RLA0_BOVIN	Large ribosomal subunit protein uL10	34.0-36.0 kDa
34.40	RPLP0	F1MDN4	F1MDN4_BOVIN	60S acidic ribosomal protein P0	34.0-36.0 kDa
35.22	TBP	Q2HJ52	TBP_BOVIN	TATA-box-binding protein	34.0-36.0 kDa
35.87	GAPDH	P10096	G3P_BOVIN	Glyceraldehyde-3-phosphate dehydrogenase	34.0-36.0 kDa
41.74	ACTB	P60712	ACTB_BOVIN	Actin, cytoplasmic 1	40.0-42.0 kDa
44.34	ACTB	A0ACH6R3D3	A0ACH6R3D3_BOVIN	Actin beta	44.0-46.0 kDa
48.04	CALR	P52193	CALR_BOVIN	Calreticulin	48.0-50.0 kDa
48.10	CALR	F6QWE4	F6QWE4_BOVIN	Calreticulin	48.0-50.0 kDa
49.86	TUBA1C	Q3ZCJ7	TBA1C_BOVIN	Tubulin alpha-1C chain	48.0-50.0 kDa
49.95	TUBB2B	Q6B856	TBB2B_BOVIN	Tubulin beta-2B chain	48.0-50.0 kDa
49.97	TUBB2B	A0ABI0P3K8	A0ABI0P3K8_BOVIN	Tubulin beta chain	48.0-50.0 kDa
50.14	TUBA1A	F1MNF8	F1MNF8_BOVIN	Tubulin alpha chain	50.0-52.0 kDa
50.43	TUBB3	Q2T9S0	TBB3_BOVIN	Tubulin beta-3 chain	50.0-52.0 kDa
50.46	TUBA1B	A0A452DIK7	A0A452DIK7_BOVIN	Tubulin alpha chain	50.0-52.0 kDa
56.03	TUBB3	A0AAF6YHE7	A0AAF6YHE7_BOVIN	Tubulin beta chain	56.0-58.0 kDa
57.27	P4HB	P05307	PDIA1_BOVIN	Protein disulfide-isomerase	56.0-58.0 kDa
59.69	ATP5F1A	F1MLB8	F1MLB8_BOVIN	ATP synthase subunit alpha	58.0-60.0 kDa
59.72	ATP5F1A	P19483	ATPA_BOVIN	ATP synthase F(1) complex subunit alpha, mitochondrial	58.0-60.0 kDa
63.35	P4HB	F6Q9Q9	F6Q9Q9_BOVIN	Protein disulfide-isomerase	62.0-64.0 kDa
67.78	CANX	G3MY89	G3MY89_BOVIN	Calnexin	66.0-68.0 kDa
69.02	HSPA1A	A0AAF6YYN3	A0AAF6YYN3_BOVIN	Heat shock protein family A (Hsp70) member 1A	68.0-70.0 kDa
69.08	LMNB1	A0AAA9TNY2	A0AAA9TNY2_BOVIN	Lamin-B1	68.0-70.0 kDa
70.26	HSPA1A	Q27975	HS71A_BOVIN	Heat shock 70 kDa protein 1A	70.0-72.0 kDa
71.24	HSPA8	P19120	HSP7C_BOVIN	Heat shock cognate 71 kDa protein	70.0-72.0 kDa
72.94	SDHA	P31039	SDHA_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
73.06	SDHA	G3MY67	G3MY67_BOVIN	Succinate dehydrogenase [ubiquinone] flavoprotein subunit, mitochondrial	72.0-74.0 kDa
74.13	LMNA	F1MYG5	F1MYG5_BOVIN	Prelamin-A/C	74.0-76.0 kDa
83.25	HSP90AB1	Q76LV1	HS90B_BOVIN	Heat shock protein HSP 90-beta	82.0-84.0 kDa
84.73	HSP90AA1	Q76LV2	HS90A_BOVIN	Heat shock protein HSP 90-alpha	84.0-86.0 kDa
89.86	TFRC	A0AAA9U1C1	A0AAA9U1C1_BOVIN	Transferrin receptor protein 1	88.0-90.0 kDa
95.37	EEF2	Q3SYU2	EF2_BOVIN	Elongation factor 2	94.0-96.0 kDa
112.64	ATP1A1	Q08DA1	AT1A1_BOVIN	Sodium/potassium-transporting ATPase subunit alpha-1	112.0-114.0 kDa
129.07	VCL	A0ABI0NNT6	A0ABI0NNT6_BOVIN	Vinculin	128.0-130.0 kDa

All molecular-weight bins are shown. Long Families and Examples / audit families cells wrap and continue across pages.

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MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
82.0-84.0 kDa	206	1	8	Apoptosis caspase/BCL; Heat shock proteins; Histone family; MAPK cascade; Mitochondrial markers	elevated	TRAF4, ZC3H14, PUM3, SLC44A5, JPH3, CRACR2A, ACAP1, , MCM5, TXLNB, MZF1, GPRIN3, PRDM9 [Histone family], QSOX1, TROP, CAPN1, LOC112449111, ORC3
84.0-86.0 kDa	202	1	5	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	PIK3R6, MID2, ADD1, AMFR, HSD17B4, NCSTN, LOXL4, SYTL5, DRC1, KIF2A, GFM1 [Mitochondrial markers], ZNF786, SATB1, ADAM11, CLPTM1L, DENND1B, UBTf,
86.0-88.0 kDa	180	0	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; Tubulin family	elevated	HSPA14 [Heat shock proteins], ELAC2, VP551, ZNF667, CHPF2, LOC112445144, ZNF184, GFM2 [Mitochondrial markers], ACAP1, LOC515755, LOC518612, PCSK7, PLCD1, RPS6KA1, HPS6, ENAH, CDHR5, SEC23A
88.0-90.0 kDa	193	1	6	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel; p53 pathway	elevated	CDH11, TPC3, ITGB5, PTPN12, FAM193B, XRRAR1, FASTKD3, MFSO6, SH3RF3, ITGB5, LOC505600, HECTD2, TTC14, TBL3, UTP14A, ITGB1, CHAMP1, NYAP1
90.0-92.0 kDa	143	0	6	Actin family; Histone family; JAK-STAT; MAPK cascade; Nuclear receptor PPAR	elevated	RINT1 [Actin family], GCFC2, CDH16, SLC9A4, , ZNF814, BOP1, ZFX, ADAM1A, KIF6, TLR2, DACT1, LRRRC8E, TLR2, PANK4, PTCO1, DLG3, PECAM1
92.0-94.0 kDa	163	0	6	Actin family; Apoptosis caspase/BCL; Heat shock proteins; Histone family; Mitochondrial markers	elevated	ATG9A, NAA35, NLGN1, DIAPH1, IQUB, OSBPL11, THSD1, ADAM23, ST14, MAP7D1, ANAPCA4, GSN, THSD1, CHSY1, MICALL1, SEMA4B, HIF1A, ZHX2
94.0-96.0 kDa	137	1	5	Actin family; Cell-cycle CDK/cyclin; Heat shock proteins; Histone family; JAK-STAT; MAPK cascade	elevated	LRRCSA, E2F8, USP16, TAX1BP1, EXD3, DNMT3B, CC2D1B, KAT2A [Histone family], EPS8L2, CDH15, TMTC2, ZC3H12B, PERM1, LZTR1, CCDC175, KLHL29, AP5Z1, TPCN1
96.0-98.0 kDa	149	0	7	Actin family; Heat shock proteins; JAK-STAT; MAPK cascade; Mitochondrial markers; NF-kB/Rel	elevated	KCNQ2, TLR4, VSIG10L, TLR4, AMPD2, GCC1, TFIIP11 [Actin family], MAP4K2 [MAPK cascade], MLXIPL [Actin family], KAZN, ADGRG1, TGFBR3, TFIIP11 [Actin family], PARP9, MVP, ABCF1, PLA2G6, TMEM245
98.0-100.0 kDa	135	0	4	Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers	elevated	TLR5, KCNQ4, ASAP3, ATP6V0A2, LRRRC8D, PIK3R5, MASTL, CSF2RB, OXR1, ZHX1, PLA2G4E, ESYT3, PDE6B, ACO1, PAN3, TRPY4, VAV1, ADAM15
100.0-102.0 kDa	126	0	5	Apoptosis caspase/BCL; Lamin family; MAPK cascade; Mitochondrial markers; Nuclear receptor PPAR; p53 pathway	elevated	PALM2, FAM111B, CHSY3, AGBL2, CBL, KIF20A, PIDD1 [p53 pathway], MAP10, DLG1, MYRFL, LEMD3, CTNNA1, ZNF629, RBM25, LOC516442, DSC2, SF3B2, SFBMT2
102.0-104.0 kDa	131	0	4	Actin family; Histone family; JAK-STAT; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	CHRD, HACE1, EXOC1, AASS [Mitochondrial markers], ESF1, LPIN2, TCAF1, TCAF1, LOC787812, TMEM8B, SYNPO2L, RFX6, USP26, GEN1, ATP2C2, , LOC618456, GTF3C3
104.0-106.0 kDa	96	0	2	Actin family; Lamin family; Mitochondrial markers; NF-kB/Rel	elevated	DTHD1, FBXO10, ICAM5, TNPO3, CCDC186, PKN1, SMCRR, WWP2, THBS3, TMTC3, CWC22, RFX1, ITIH5, EXTL3, NSMAF, EDEM3, ITIH5, ERCC4
106.0-108.0 kDa	103	0	3	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; Mitochondrial markers; Tubulin family	elevated	TRAK1, IL6ST, ADGRD2, BCLAF1 [Apoptosis caspase/BCL], CD163L1, ADGRF3, MEIOC, PITPNM3, RB1, CFAP69, CUL4B, PRPF40A, CASD1, RNF216, TUBGCP2 [Tubulin family], ANO6, PPP4R1, WDR3
108.0-110.0 kDa	106	0	3	Actin family; Cell-cycle CDK/cyclin; Histone family; MAPK cascade	elevated	EMILIN1, INTS4, SLC8A1, INTS5, TMEM63A, VAV3, MAP3K13 [MAPK cascade], IPO13, PAM, PSD3, MAP3K13 [MAPK cascade], GABBR1, PAM, CCDC191, KANK4, PTPRN2, GNAS, NEK9
110.0-112.0 kDa	89	0	2	Actin family; Nuclear receptor PPAR; p53 pathway	elevated	RAI14, KIF5C, CACNA2D4, ZSWIM4, MIB1, FGD1, VPS18, LATS2, GTF2I, SMG8, EPHA3, ANKRD35, PEAR1, STOX1, CENPF, RNF111, DD12, UBA7
112.0-114.0 kDa	94	1	0	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers	elevated	EPHA7, ZNF628, EML4, SEZ6L, ATP1A2, IPO11, DSG1, SLC4A5, TRPC4, MAN2C1, CORIN, DSG1, CBLB, MAP1S, LOC528518, MAP1S, C25H16orf96, PPFIBP1
114.0-116.0 kDa	84	0	6	Actin family; Apoptosis caspase/BCL; Heat shock proteins; MAPK cascade; Mitochondrial markers; Tubulin family	elevated	RASIP1 [Actin family], SORBS2, EPHA5, PKN, HERC6, ZNF154, LOC112442305, NLRP9, SEMA6A, LIG3, NLRP9, ATP4A, LOC107132586, B4GALNT3, CNTN6 [Actin family], ST18, PUM2, FARP1
116.0-118.0 kDa	90	0	1	Actin family; Apoptosis caspase/BCL; Histone family; Mitochondrial markers	elevated	TMEM132E, STK31, TAOK1, LOC100297905, EMC1, ITGA9, USP38, ZMIZ1, HIP1R [Actin family], KCNH3, CORO7, GOLGA2, CSF1R, WDR49, ITGA9, ABL2, ANKRD28, RIMBP2
118.0-120.0 kDa	68	0	2	Actin family; Histone family	routine	MED24, ZNF445, RPS6KC1, IDE, PTK7, IDE, PPP1R13B, FBRSL1, NLRP3, CPED1, SGSM2, ZNF516, U2SURP, NLRP12, LOC107131271, MRTFA, PLEKH45,
120.0-122.0 kDa	87	0	2	AKT pathway; Actin family; Apoptosis caspase/BCL; MAPK cascade	elevated	ROR2, FBXW10B, SEC24A, MIS18BP1, SYNPO, FAM120C, USPL1, TNK2, SEMA5A, GUCY2D, TMEM132C, GEMIN4, RANBP2, CFAP251, SUPT5H, GUCY2D, DNA2, ADAMTS17
122.0-124.0 kDa	73	0	2	JAK-STAT; Lamin family; Tubulin family	elevated	MEGF10, BNC2, WDR6, SEMA5B, CEBP2, STARDB, PLCL1, TUBGCP5 [Tubulin family], OTUD4, PLD2, AMOT, SREBF2, TRIM33, , FAM184B, UBE4A, KIF1C, GARNL3
124.0-126.0 kDa	85	0	1	Actin family; Tubulin family; p53 pathway	routine	SRGAP1, PLD1, SPECC1L, GRIP1 [Actin family], AP3B2, , AFF4, ATN1, TMC3, ESYT1, FMNL2, R3HDM1, GUCY2F, USP8, GUCY2F, UPF1, AGAP2, PDE3B
126.0-128.0 kDa	46	0	2	Actin family; Histone family; Mitochondrial markers	routine	CFAP70, PKP4, LAT51, DAAM1, CILP2, PCDH17, RFC1, NR1P1 [Actin family], SLC12A7, MAGEL2, MCM9, OTOA, PIK3CG, PTPN21, HDAC5 [Histone family], DENND2B, CCDC150, TCERG1
128.0-130.0 kDa	69	1	2	Actin family; Lamin family; Mitochondrial markers	elevated	AP4E1, RANBP17, FNIP1 [Actin family], ERC1 [Actin family], RBL2, SIN3B, TRAPPC11, TRAPPC11, PDZD8, PLEKHG4, RBM33, LARP1, USP7, POLR3B, ARHGAP39, HELB, TSC1, TBC1D8B
130.0-132.0 kDa	63	0	2	Actin family; JAK-STAT; Lamin family	elevated	PUM1, ATP8A1, CNMA1, ADCY6, PPP2R3A, HIPK2, HPS5, KIAA1755, PROM10, ADAR, LOC536660, ARID5B, SMARCA1 [Actin family], FAM184A, ZNF862, KCNT2, ZSCAN20, SLC12A2
132.0-134.0 kDa	73	0	3	Actin family; Apoptosis caspase/BCL; Histone family; JAK-STAT; NF-kB/Rel; Tubulin family	elevated	COBLL1, CILP, TTF2, TBC1D8, CATSPERG, TBC1D1, DAB2IP [Actin family], NOL5, RBM20, NRCAM, WWC2, NEK10, TECPR1, PALB2, LEPR, WNK4, NCAPG2, VWA5B1
134.0-136.0 kDa	58	0	1	Actin family; Lamin family; MAPK cascade	routine	CTC1, EGFR, LOC515670, DGKD, IGDC4, LOC516849, ATP11B, C23H6orf132, PLCB2, SEC31A, FILIP1 [Actin family], Lamin family], CRB2, MN1, RTEL1, SPATA31G1, MAPKBP1 [MAPK cascade], CAMSAP3, ADAMTS18
136.0-138.0 kDa	53	0	0	Actin family; NF-kB/Rel	routine	NID1, PCDH9, CGN, EHBP1, PLCB4, LOC100847574, ANO8, INF2, CHL1, UNC13B, CAND1, AP3D1, WASHC4, MYBPC1, TAF2, SLC4A2, ZNF592, PCDH7
138.0-140.0 kDa	59	0	2	Actin family; Cell-cycle CDK/cyclin; Histone family	routine	GAK [Cell-cycle CDK/cyclin], TTC17, POLRMT, AUTS2, ITPRID2 [Actin family], PTPRO, , PPPIA4 [Actin family], FBF1, FNDCA3, TAOX2, RRGRI1 [Actin family], COPA, TPP2, CEACAM20, ATP13A3, WDR11, RRGRI1 [Actin family]
140.0-142.0 kDa	57	0	3	Actin family; Cell-cycle CDK/cyclin; Histone family; p53 pathway	elevated	ADCY8, TIMELESS, KIF4A, MTERF4, JCAD, FAM120A, CRAMP1, WDR35, CFH, RTEL1, UBE2O, VARS1, CFH, DHX38, NEK1, MTR, MTR, ERCC6L
142.0-144.0 kDa	41	0	1	Actin family; Histone family	routine	TNKS, SORBS1, PPPIA2 [Actin family], PER2, TRPS1, MMS22L, ZFAT, ADGRA2, TNKS2, ANKS1B, COBL, KCNT1, RTL9, PRX, UMODL1, ADGRA3, NID2, IRS4
144.0-146.0 kDa	38	0	1	Actin family; MAPK cascade	routine	SIK3, VWA3B, UBN2, STAG1, REXO1, INSRR, MROH7, AH1, TBC1D32, ZNF541, ANKFN1, ARID4A, CNTNAP4 [Actin family], OVCH1, COL5A2, MAP3K6 [MAPK cascade], DENND5B, RPTN
146.0-148.0 kDa	47	0	2	Actin family; Lamin family; MAPK cascade	routine	WDR17, SPAG9 [Actin family], SPAG5, ZNF532, STK36, RGS22, SH3TC1, PNPLA7, BMS1, IARS1, PITPNM2, ABCB11, USP42, NPC1L1, XDH, ARID4B, MST1R, ERBB4
148.0-150.0 kDa	49	0	3	Actin family; Apoptosis caspase/BCL; Cell-cycle CDK/cyclin; MAPK cascade	elevated	ABCC4, ROPOR1 [Actin family], LOC528412, AOX4, ZNF423, DENND3, AOX2, AATK, FAM83H, RNF123, PLCG1, FAM205A, TUT1, PLEKHG3, LOC509854, CCNB3 [Cell-cycle CDK/cyclin], TDRD12, SYNGAP1
150.0-152.0 kDa	45	0	0	Histone family	routine	ACE, CECR2, CARMIL3, RGS12, PHLPP2, COL17A1, ARHGEF18, TONSIL, TTC41, USP19, SBN02, ACIN1, MTMR4, PDS5A, PNPLA6, CASKIN1, CLUH, SOS1
152.0-154.0 kDa	32	0	1	Actin family; MAPK cascade	routine	MYBBP1A, UBE4B, ZFYVE9, ZSWIM8, BRD4, PLEKH1H, TNS2, CPD, MYO3B, ABCC12, CEMIP, CYFIP2 [Actin family], MSH6, PLCI2, PLCG1, ATRNL1 [Actin family], SGO2, ZMYM3
154.0-156.0 kDa	29	0	0	Histone family	routine	ZMYM2, CNGB1, AKNA, TONSL, , ABCC11, SBN01, HEG1, CADPS, CEMIP2, L1CAM, LMTK3, KIAA0232, WC1, IGF1R, FAM135B, MET, INSR
156.0-158.0 kDa	26	0	1	Actin family; Apoptosis caspase/BCL; Histone family	routine	RAB3GAP2, ATAD2, CNTNAP1 [Actin family], NCAPD2, BCL9L [Apoptosis caspase/BCL], LOC132344466 [Histone family], IGFSF9B, NKTR, POGZ, WC1.3, NCOA1, FMN1, ANKRD50, MED23, GAPVD1, IBTK, RFX7, LOC100337390
158.0-160.0 kDa	22	0	1	Cell-cycle CDK/cyclin; Histone family; Mitochondrial markers	routine	SCAPER [Cell-cycle CDK/cyclin], PRG4, DCC, NPHP4, TECPR2, ROCK1, ABCC10, DMBT1, DROSHA, MAGI2, NCOA2, RIC1, LRRPCK [Mitochondrial markers], BLTP3A, NLRP1, NEO1, SYDE2, WRN
160.0-162.0 kDa	40	0	2	Actin family; Histone family	routine	RASAL2, COL22A1, TDRD9, BLM, YTHDC2, MAGI1, VPSB, FANCA, COL4A1, COL4A3, ATP10D, MED14, TRAPPC8, ROCK2, FGD6, KIF7, ATP7B, ZNF226
162.0-164.0 kDa	30	0	1	Histone family; MAPK cascade	routine	DOT1L [Histone family], , ADGRL1, SCAF8, A2ML1, PTPRU, COL4A6, PLB1, UACA, FAM193A, PTPRT, CR2, COL11A2, MYOM3, SCAF11, RAD54L2, ERBIN, FLT1
164.0-166.0 kDa	34	0	2	Cell-cycle CDK/cyclin	routine	CDK12 [Cell-cycle CDK/cyclin], COL4A4, PXDN, FYCO1, PLEKHG1, COL16A1, NRXN1, ATP8B3, PTPRK, IRAG2, PDS5B, CDK13 [Cell-cycle CDK/cyclin], GRIN2A, ATAD2B, , PRTG, FGD5, ABCC6
166.0-168.0 kDa	30	0	1	Mitochondrial markers	routine	PHLPP1, MRC2, LOC506828, ZNF518A, NAIP, RALGAPB, RRBP1, GRIN2B, CPS1 [Mitochondrial markers], NISCH, LOC508459, ROCK2, NEURL4, RSF1, SYCP2, NCAPD3, CUX1, NFAF5
168.0-170.0 kDa	24	0	0	Actin family	routine	, IQSEC2, , POLA1, PLEKH2, GEMINS, PLA2R1, MYOM2, ZNF831, PLIN4, DCAF1, FAM135A, DIP2A [Actin family], CLASP1, ZFYVE16, EIF3A, HFM1, URB2
170.0-172.0 kDa	24	0	1	AKT pathway; Lamin family	routine	RERE, NHLN1, SHROOM2, FHAD1, PIK3C2G, ARHGAP35, BAZ1B [AKT pathway], ARAP3, EPRS1, TRPM2, MAP4, AQR, MAST1, SHROOM4, DISP1, LAMC3 [Lamin family], TUT7, ADGRB3
172.0-174.0 kDa	26	0	1	Histone family; MAPK cascade; Nuclear receptor PPAR	routine	COL5A3, COL11A2, ARHGAP5, LRRC7, ADGRB1, PCF11, ARHGEF12, TULP4, KIAA0586, EFCAB6, MAP3K19 [MAPK cascade], SCRIB, NWD1, PPRC1 [Nuclear receptor PPAR], RIMBP3B, GLI3, PHRF1, ZMYM6
174.0-176.0 kDa	16	0	0	Actin family	routine	AGL, , ABCC9, TOP2A, TMEM131L, TMEM131L, ATRN [Actin family], COL24A1, SETBP1, ERCC6L2, UGGT2, DUOX2, CEP170B, KDM5C, BAZ1A, CFAP74
176.0-178.0 kDa	20	0	0	Actin family; Lamin family	routine	ADGRL3, ALK, SMG6, THSD7B, ERCC6L2, PTPN23, ROBO1, KDM6B, HECW2, EIF4G1, PREX2, ADGRB2, SHOC1, ABCC8, DIP2C [Actin family], HECW1, DUOX1, LAMC1 [Lamin family]
178.0-180.0 kDa	15	0	1	Actin family; MAPK cascade; Tubulin family	routine	DNMBP, MEGF6, ABCA9, ADAMTS12, SYNM, TUBGCP6 [Tubulin family], PLXNC1, SMARCA2 [Actin family], UGGT1, TNS3, NOS1, MAP3K4 [MAPK cascade], COL18A1, DIP2B [Actin family], ARHGEF5
180.0-182.0 kDa	18	0	1	Actin family	routine	, KATNIP [Actin family], LRP5, DEPDC5, SMARCA4 [Actin family], MROH2B, PAPP4, KIF14, COL11A1, TARBP1, KIF21B, MYO3A, USP32, DENND4B, TOP2B, IQGAP2, RB1CC1, RIMS2
182.0-184.0 kDa	19	0	0	Actin family	routine	PRDM2, CEP170, MROH1, SMARCA4 [Actin family], PTPRV, WDR62, RAPGEF6, DNMT1, ALPK3, MADD, ADAMTS7, MTL2L, ALS2, TNKS1BP1, THSD7A, UNC13A, ABCA10, TCHH
184.0-186.0 kDa	14	0	1	Histone family	routine	MBD5, PIK3C2B, ABCA6, ADAMTSL3, LRP6, COL5A1, WIZ, MYOM1, PREX1, COL27A1, LTBP1, TOPAZ1, SETD1A [Histone family], AKAP12

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
186.0-188.0 kDa	18	0	0	Lamin family	routine	ABCA5, CAMSAP1, THOC2, LOC528040, ABCA16, SPHKAP, TRPM1, LAMB4 [Lamin family], RREB1, EIF2AK4, ADCY10, USP54, RIMS1, C3, CC2D2A, KIF21A, TUT4, TIAM2
188.0-190.0 kDa	12	0	0	Actin family	routine	RNF17, ABCA14, PLCH1, C5, CTTNBP2 [Actin family], KIF26A, KIAA1210, IQGAP1, SLX4, C4orf54, SIPA1L2, KCP
190.0-192.0 kDa	14	0	1	Actin family	routine	BCORL1, FRMPD3, CCDC180, MROH2A, MON2, PIK3C2A, ITSN2, SHANK3, CUL7, STRC, BCOR, CLTC, PKD1L3 [Actin family], KDMA5
192.0-194.0 kDa	25	0	0		routine	MROH1, MARF1, KDM3B, ABCA3, PEAK1, IQGAP3, PBRM1, TET3, C4, NRXN2, COL14A1, CASZ1, CFAP43, LOC107131209, SCN7A, XRN1, LRRIQ1, C4A
194.0-196.0 kDa	15	0	0	Actin family	routine	POLR1A, SIPA1L3, ARAP2, DLEC1, RICTOR, XIRP1 [Actin family], FRMPD4, MYO5B, TNRC6C, ITSN1, TRPM3, GCC2, CDC42BPA, NUP98, NES
196.0-198.0 kDa	20	0	0	Actin family; Lamin family	routine	LAMB2 [Lamin family], KIDINS220 [Actin family], NUP188, SKIC3, CEP152, ANKRD26, CHD1, NRAP, IFT172, ARHGEF28, TSP0AP1, NWD2, SIPA1L1, QRICH2, LAMB1 [Lamin family], ZC3H13, MAST2, PAPPA2
198.0-200.0 kDa	8	0	0		routine	LTBP2, N4BP2, ARFGEF2, LY75, ZNF646, SUPT6H, BTBD8, RAI1
200.0-202.0 kDa	16	0	1		routine	TSC2, TANC1, NALCN, CDCDC1, NAV1, TOGARAM1, ITGB4, UBR2, UBR1, FOCAD, KIF13A, ADGB, RBBP6, PRR12, QSER1, KIF1A
202.0-204.0 kDa	13	0	0	Lamin family; p53 pathway	routine	TNS1, CPAMD8, LMO7, LAMA4 [Lamin family], SHPRH, PARP4, PARP14, MYO5C, ECPAS, MED12L, CRYBG2, NUP210 [p53 pathway], BRWD3
204.0-206.0 kDa	11	0	0	Actin family; Histone family	routine	ZNF142, LTN1, TMEM131, SHANK2, KIF1B, KIF26B, PAT3, KMT2E [Histone family], PHIP [Actin family], NCOA6, NLRC5
206.0-208.0 kDa	18	0	0	Actin family	routine	WDR90, AFDN, TJP1, BRCA1, BRCA1, SCN11A, FNDCl, PCSK5, MDC1, BTAF1, MICAL2 [Actin family], EIF4G3, SCN4A, ANK1, DBSL1, VIRMA, SBF1, PLXNB3
208.0-210.0 kDa	18	0	1	Actin family; Cell-cycle CDK/cyclin; p53 pathway	routine	SBF2, NOTCH4, KIAA1549, PLXNA3, SI, PDCD11, CHD2, PDCD11, TNRC6A, ARFGEF1, ZNF236, TICRR [Actin family], EFCAB3, NUP210L [p53 pathway], MTCL1, HCFCl, RALGAPA2, CDK5RAP2 [Cell-cycle CDK/cyclin]
210.0-212.0 kDa	16	0	0		routine	AKAP11, VWA8, FER1L6, PLXNA2, GBF1, NYNRIN, CACNA1S, KIAA1217, PTPRF, LCOLR, ZNF106, PSME4, C2CD6, MIA3, LTBP2, DOCK2
212.0-214.0 kDa	18	0	1	Actin family; Histone family; p53 pathway	routine	PLXND1, LOC101902149, SETD1B [Histone family], MCM3AP, KIF20B, ATG2A, UBR3, WDR81, TRPM7, PLXNA4, SPICE1, TP53BP1 [p53 pathway], NCKAP5, BAZZA, ATAD5, GREB1L, ARHGEF4, MPRIP [Actin family]
214.0-216.0 kDa	15	0	0		routine	MIA3, DENND4A, ARFGEF1, PLXNA1, PTPRD, TTC6, PPL, PLXNB2, DOCK5, SHROOM3, LRP4, GREB1, ADAMTS20, DENND4C, PCDH15
216.0-218.0 kDa	15	0	0	Actin family; p53 pathway	routine	ADAMTS9, ANAPC1, MAP2, ANKRD31, TAF1, AGRN, ARHGAP21 [p53 pathway], POLR2A, HELZ, ASPM, RLF, MICAL3 [Actin family], EML6, THADA, DOCK1
218.0-220.0 kDa	10	0	0	p53 pathway	routine	DICER1, MYO5A, DICER1, MPDZ, CRYBG1, NUP214 [p53 pathway], LCT, CACNA1F, EML5, ZNFX1
220.0-222.0 kDa	13	0	1	Apoptosis caspase/BCL	routine	TRIP12, BRD10, AK9, CHD4, ARHGEF17, PTPRS, CASP8AP2 [Apoptosis caspase/BCL], SDK1, MYH7B, MYH15, TDRD15, HEATR5A, GSE1
222.0-224.0 kDa	18	0	0		routine	KIF13B, DSCAM, SCN10A, MYH4, CROCC, PCNX3, MYH8, CHD5, MYH1, TCF20, MYH7, MYH7, MYH2, MYH2, MYH13, DLG5, MYH3, MYH6
224.0-226.0 kDa	10	0	0	Histone family	routine	SCN2A, HEATR5B, SCN9A, LRRK1, CKAP5, SCN8A, KAT6A [Histone family], PTPN13, TTC3, SMCHD1
226.0-228.0 kDa	9	0	0		routine	TET2, ZFC3H1, SHANK1, PHF3, OTOF, PCNX2, TRIP11, MYH9, SCN2A
228.0-230.0 kDa	12	0	0	Histone family	routine	SCN5A, MYH11, TANC2, ALPK2, SCN1A, KIAA1549L, KAT6B [Histone family], MYH10, WNK2, DOCK6, PRRC2A, EVPL
230.0-232.0 kDa	6	0	0		routine	NUP205, FANCM, DSCAML1, ESPL1, ARHGAP32, TRPM6
232.0-234.0 kDa	9	0	0		routine	ICE1, PLXNB1, CCDC88A, MYOF, ATG2B, CHD3, ANKRD12, DOCK4, RP1
234.0-236.0 kDa	13	0	0		routine	LOC101904413, ZNF638, MYLK, ABCA7, PI4KA, CABIN1, TDRD6, RP1L1, LOXHD1, MYO10, MYO10, RP1, PI4KA
236.0-238.0 kDa	7	0	0	Actin family	routine	GTF3C1, CIT [Actin family], , LOC104972622, USF3, D1D01, DOCK11
238.0-240.0 kDa	12	0	0		routine	MYH14, PIKFYVE, CACNA1C, DYSF, TET1, FER1L5, MED13, F5, ARMCX4, TECTA, APC2, ZNF407
240.0-242.0 kDa	10	0	0		routine	ARFGEF3, DYSF, TNC, SPEF2, SDK2, YLPM1, CACNA1I, ASXL3, MYO7B, DOCK8
242.0-244.0 kDa	12	0	0		routine	ARID1A, NBAS, MED13L, HEATR1, DOCK7, ZGRF1, DOCK9, INTS1, CAD, PCM1, MED12, MYO9B
244.0-246.0 kDa	6	0	0		routine	SNRNP200, NOTCH3, SVIL, FREM1, BAZ2B, CACNA1D
246.0-248.0 kDa	9	0	0		routine	MALRD1, SEC16A, ACAN, GON4L, ACAN, COL6A6, DOCK3, RTTN, WNK1
248.0-250.0 kDa	3	0	0		routine	PTPRB, F5, NIN
250.0-252.0 kDa	9	0	0		routine	UNC13C, CSPG4, KNTC1, ascc3, SORL1, MYO7A, PKDREJ, DOCK10, CLIP1
252.0-254.0 kDa	4	0	0		routine	SPAG17, KIAA0100, ARID1B, AHCTF1
254.0-256.0 kDa	7	0	0	Heat shock proteins	routine	ABCA1, URB1, DNAJC13 [Heat shock proteins], CACNA1E, PTPRZ1, NAV3, DOP1B
256.0-258.0 kDa	8	0	0		routine	ZNF318, AKAP6, PTPRQ, CIC, PLCE1, ABCA4, HIVEP3, PCNX1
258.0-260.0 kDa	1	0	0		routine	C2CD3
260.0-262.0 kDa	3	0	0		routine	NAV2, CACNA1B, LOC101903126
262.0-264.0 kDa	9	0	0	Histone family	routine	ROS1, POLE, SON, GPR179, BRWD1, CACNA1G, F8, KNL1, EP300 [Histone family]
264.0-266.0 kDa	6	0	0	Histone family	routine	PEG3, OTOGL, PEG3, CREBBP [Histone family], ACACA, NOTCH2
266.0-268.0 kDa	2	0	0		routine	HIVEP2, CNOT1
268.0-270.0 kDa	4	0	0		routine	SPTB, CNTRL, NOTCH1, MAP1B
270.0-272.0 kDa	5	0	0		routine	SPTBN2, ACACA, TTC28, TLN1, NACAD
272.0-274.0 kDa	10	0	0		routine	FN1, ZFHX2, TLN2, ANKHD1, ACACB, CEP192, PRPF8, NCOR2, NCOR1, GOLGA4
274.0-276.0 kDa	8	0	0		routine	MYO18A, SPTBN1, IGF2R, IGF2R, FASN, TPR, FASN, ANKRD17
276.0-278.0 kDa	4	0	0		routine	CEP250, DOP1A, STAB2, COL6A5
278.0-280.0 kDa	2	0	0	Lamin family	routine	FLNB [Lamin family], STAB1
280.0-282.0 kDa	4	0	0		routine	MYO18B, ZFYVE26, CUL9, SPTA1
282.0-284.0 kDa	6	0	0	Actin family	routine	RIF1, CACNA1A, SPG11, PDE4DIP [Actin family], PIEZO1, JMJD1C
284.0-286.0 kDa	5	0	0		routine	BDP1, SPTAN1, FN1, RALGAPA1, HIVEP1
286.0-288.0 kDa	5	0	0	Histone family; Lamin family	routine	LRRK2, SETD2 [Histone family], FLNC [Lamin family], ATRX, IGSF10
288.0-290.0 kDa	4	0	0		routine	BAHCC1, SPTBN4, HECTD1, POLQ
290.0-292.0 kDa	5	0	1		routine	ZNF462, MTOR, ANKRD11, USP9X, CHD8
292.0-294.0 kDa	8	0	0	Histone family	routine	EPG5, GCN1, USP9Y, MYO9A, ABCA12, COL7A1, KMT2B [Histone family], CFAP46
294.0-296.0 kDa	7	0	0		routine	USP24, ZAN, TACC2, TEPI, PDZD2, TASOR2, CEP295
296.0-298.0 kDa	4	0	0	Histone family	routine	NSD1 [Histone family], MEGF8, SRRM2, UNC79
300.0-302.0 kDa	2	0	0		routine	MGAM, TENM3
302.0-304.0 kDa	9	0	0		routine	ATR, ABCA2, MXRA5, ITPR3, TG, SETX, TG, ZNF292, ITPR3
304.0-306.0 kDa	4	0	0		routine	FBN3, CHD6, AKAP13, TENM1
306.0-308.0 kDa	2	0	0		routine	TENM2, ITPR2
308.0-310.0 kDa	6	0	0		routine	ITPR2, ITPR1, TEX15, UBR5, CENPE, NBEAL1
310.0-312.0 kDa	3	0	0		routine	VWF, PRUNE2,
312.0-314.0 kDa	4	0	0		routine	FBN1, OTOG, TNRC18, MXRA5
314.0-316.0 kDa	5	0	0		routine	APC, FBN2, LOC616782, TENM4, PRRC2C
316.0-318.0 kDa	2	0	0		routine	NIPBL, LRBA
318.0-320.0 kDa	4	0	0		routine	UTP20, FCGBP, BOD1L1, CRYBG3
322.0-324.0 kDa	1	0	0		routine	CEP290
324.0-326.0 kDa	3	0	0	Histone family	routine	CELSR2, PIEZO2, ASH1L [Histone family]
326.0-328.0 kDa	2	0	0		routine	CHD9, NBEA
330.0-332.0 kDa	1	0	0		routine	CELSR1
332.0-334.0 kDa	3	0	0		routine	CHD7, DSP, COL12A1
334.0-336.0 kDa	4	0	1		routine	SRCAP, NF1, ZZEF1, SYNE1
336.0-338.0 kDa	2	0	0		routine	EP400, LAMA1

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
338.0-340.0 kDa	5	0	0		routine	MAP1A, BPTF, FRY, KALRN, COL6A3
340.0-342.0 kDa	4	0	0		routine	DMXL2, TRANK1, MGA, DMXL1
342.0-344.0 kDa	2	0	0	Lamin family	routine	IQCN, LAMA2 [Lamin family]
344.0-346.0 kDa	2	0	0		routine	HTT, RANBP2
346.0-348.0 kDa	2	0	0		routine	DCHS1, TRIO
348.0-350.0 kDa	1	0	0		routine	WDR87
350.0-352.0 kDa	4	0	0		routine	ATM, CEP350, FREM2, FRYL
352.0-354.0 kDa	3	0	0		routine	CFAP54, REV3L, WDFY4
354.0-356.0 kDa	2	0	0		routine	SPEG, LOC101902689
356.0-358.0 kDa	1	0	0		routine	CFAP47
358.0-360.0 kDa	4	0	0		routine	MAST4, CELSR3, PCNT, CDH23
360.0-362.0 kDa	2	0	0		routine	VPS13A, SZT2
362.0-364.0 kDa	1	0	0		routine	UNC80
364.0-366.0 kDa	1	0	0		routine	CPLANE1
366.0-368.0 kDa	2	0	0	Lamin family	routine	LAMA3 [Lamin family], LOC539323
368.0-370.0 kDa	1	0	0		routine	VCAN
370.0-372.0 kDa	1	0	0		routine	MKI67
374.0-376.0 kDa	2	0	0		routine	CENPF, EPPK1
376.0-378.0 kDa	1	0	0		routine	GOLGB1
384.0-386.0 kDa	1	0	0		routine	BRCA2
386.0-388.0 kDa	1	0	0		routine	CSMD1
388.0-390.0 kDa	2	0	1	NF-kB/Rel	routine	SVEP1, RELN [NF-kB/Rel]
392.0-394.0 kDa	1	0	0		routine	WDFY3
394.0-396.0 kDa	4	0	0		routine	UTRN, ASPM, CSMD2, ZFHx4
396.0-398.0 kDa	2	0	0		routine	CUBN, ZNF469
398.0-400.0 kDa	1	0	0	Lamin family	routine	LAMA5 [Lamin family]
400.0-402.0 kDa	1	0	0	Actin family	routine	SPEN [Actin family]
402.0-404.0 kDa	2	0	0		routine	ZFHx3, USP34
404.0-406.0 kDa	2	0	0	Actin family	routine	XIRP2 [Actin family], CSMD3
410.0-412.0 kDa	1	0	0		routine	SMG1
414.0-416.0 kDa	1	0	0		routine	BSN
418.0-420.0 kDa	1	0	0		routine	VPS13C
428.0-430.0 kDa	1	0	0		routine	LYST
434.0-436.0 kDa	2	0	0	Histone family	routine	LYST, KMT2A [Histone family]
436.0-438.0 kDa	2	0	0		routine	TRRAP, VPS13B
440.0-442.0 kDa	2	0	0		routine	CMYA5, PKHD1
442.0-444.0 kDa	1	0	0		routine	FRAS1
452.0-454.0 kDa	1	0	0		routine	DNAH12
458.0-460.0 kDa	1	0	0		routine	ANK2
460.0-462.0 kDa	3	0	0	Actin family	routine	PKD1 [Actin family], DNAH7, MUC19
464.0-466.0 kDa	1	0	1		routine	PRKDC
466.0-468.0 kDa	3	0	0		routine	AKAP9, DNAH3, PKHD1L1
468.0-470.0 kDa	1	0	0	Heat shock proteins	routine	HSPG2 [Heat shock proteins]
472.0-474.0 kDa	1	0	0		routine	TNXB
474.0-476.0 kDa	1	0	0		routine	DNAH6
478.0-480.0 kDa	2	0	0		routine	ANK3, FAT2
482.0-484.0 kDa	4	0	0		routine	HUWE1, ALMS1, STARD9, MUC5B
484.0-486.0 kDa	1	0	0		routine	HECTD4
486.0-488.0 kDa	1	0	0		routine	DNAH1
490.0-492.0 kDa	1	0	0		routine	VPS13D
492.0-494.0 kDa	1	0	0		routine	DYNC2H1
498.0-500.0 kDa	1	0	0		routine	FAT3
504.0-506.0 kDa	1	0	0		routine	LRP1
506.0-508.0 kDa	1	0	0		routine	FAT1
508.0-510.0 kDa	1	0	0		routine	DNAH2
510.0-512.0 kDa	1	0	0		routine	DNAH9
514.0-516.0 kDa	3	0	0		routine	, DNAH8, APOB
518.0-520.0 kDa	2	0	0		routine	SACS, LRP2
520.0-522.0 kDa	1	0	0		routine	MYCBP2
526.0-528.0 kDa	1	0	0		routine	HERC2
530.0-532.0 kDa	2	0	0		routine	DNAH14, PLEC
532.0-534.0 kDa	2	0	0		routine	HERC1, BIRC6
534.0-536.0 kDa	2	0	0		routine	DYNC1H1, DNAH5
536.0-538.0 kDa	2	0	0		routine	DNAH11, PCLO
542.0-544.0 kDa	4	0	0	Histone family	routine	KMT2C [Histone family], FAT4, SSPO, SSPO
544.0-546.0 kDa	1	0	0		routine	HMCN2
552.0-554.0 kDa	1	0	0		routine	RYR3
554.0-556.0 kDa	1	0	0		routine	BLTP1
556.0-558.0 kDa	1	0	0		routine	ABCA13
564.0-566.0 kDa	2	0	0		routine	RYR2, RYR1
576.0-578.0 kDa	1	0	0		routine	USH2A
578.0-580.0 kDa	1	0	0		routine	UBR4
592.0-594.0 kDa	1	0	0		routine	AHNAK2
594.0-596.0 kDa	1	0	0		routine	LOC512869
600.0-602.0 kDa	1	0	0		routine	AHNAK
602.0-604.0 kDa	1	0	0	Histone family	routine	KMT2D [Histone family]
612.0-614.0 kDa	1	0	0		routine	HMCN1
636.0-638.0 kDa	1	0	0		routine	MDN1
688.0-690.0 kDa	1	0	0		routine	ADGRV1
760.0-762.0 kDa	1	0	0	Actin family	routine	FSIP2 [Actin family]
772.0-774.0 kDa	1	0	0		routine	NEB
796.0-798.0 kDa	1	0	0		routine	SYNE2

MW bin	N	Markers	Signal	Families	Priority	Examples / audit families
854.0-856.0 kDa	1	0	0	Actin family	routine	MACF1 [Actin family]
862.0-864.0 kDa	1	0	0		routine	DST

Proteins sorted from lowest to highest molecular weight

The PDF shows the first 120 proteins. The complete table is in proteome_wb_integrity_atlas.csv.

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
0.97	MGMT	P29177	Methylated-DNA--protein-cysteine methyltransferase (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.08		P15507	Morphine-modulating neuropeptide B	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.10		A0AAA9S596	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.14		P69206	Morphogenetic neuropeptide	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.38		A0AAA9S118	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.42	HOOK3	P60316	Protein Hook homolog 3 (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.48		A0AAA9TPS1	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.58		A0AAA9TW91	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.84		A0AAA9T289	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.92		P15506	Morphine-modulating neuropeptide A	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.93		A0ABI0P1B4	Uncharacterized protein	0.0-2.0 kDa		standard full-blot and antibody-validation checks
1.95		P22779	Cytochrome P450 2A (Fragment)	0.0-2.0 kDa		standard full-blot and antibody-validation checks
2.02		A0AAA9TEG0	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.13		A0A3Q1N275	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.18		A0ABI0NYC6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0AAA9SIV3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0AAA9SLK5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.20		A0A3Q1LQI6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.22		A0AAA9TM45	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.23		A0A3Q1MRH6	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.24		A0AAA9RUI4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.26		A0AAA9S011	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.30		A0AAA9TFU1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.32		A0AAA9S6L4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45		A0AAA9RTD3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.45		A0AAA9U0W5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.49		A0ABI0NPJ1	Ribosomal protein L32	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.58		A0AAA9RZ15	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.59		A0AAA9TVY5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.63		A0A3Q1M3P4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.74		A0ABI0P503	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.77		A0A3Q1M0Q7	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.78		A0AAA9TXY4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.81		A0ABI0P2W9	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.87		A0AAA9SLS3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.95		A0AAA9SUM1	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.95		A0AAA9SIM7	LASP1 neighbor	2.0-4.0 kDa		standard full-blot and antibody-validation checks
2.95		A0AAA9RXK3	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.01		A0ABI0NQN8	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.06	SCT	P63296	Secretin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.08		A0AAA9TF52	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.13	DEFB	F6RMW4	Defensin, beta	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.16		A0A3Q1LY62	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.23		A0AAA9TYS4	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.24	SMIM47	A0AAA9TSQ3	Small integral membrane protein 47	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.31	COG5	P83437	Conserved oligomeric Golgi complex subunit 5	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.34		A0ABI0P650	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.49		A0AAA9TJP5	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.60	CALCA	P01260	Calcitonin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.65		A0A3Q1MX44	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.73		A0AAA9T609	Uncharacterized protein	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.76	HRURF	A0AAA9T445	HR upstream open reading frame	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.77	SLN	A0A3Q1MB76	Sarcophilin	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.88	STRIT1	A0ABI0PID4	Small transmembrane regulator of ion transport 1	2.0-4.0 kDa		standard full-blot and antibody-validation checks
3.92	BLACAT1	A0AAA9TV63	BLACAT1 overlapping LEMD1 locus	2.0-4.0 kDa		standard full-blot and antibody-validation checks
4.03		A0ABI0NT89	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.08		A0ABI0P616	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.11	DEFB12	P46170	Beta-defensin 12	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.19	OST4	A0AAA9T743	Oligosaccharyltransferase complex subunit 4, non-catalytic	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.28	DEFB1	P46159	Beta-defensin 1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.36	DEFB8	P46166	Beta-defensin 8	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.45	DEFB13	P46171	Beta-defensin 13	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.47		A0AAA9S0W4	SPTY2D1 opposite strand	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.49		A0AAA9S347	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.64		A0A3Q1N0R6	Chromodomain helicase DNA binding protein 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.65	DEFB2	P46160	Beta-defensin 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.81	TMSB10	P21752	Thymosin beta-10	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.84	DEFB6	P46164	Beta-defensin 6	4.0-6.0 kDa		standard full-blot and antibody-validation checks
4.90		A0AAA9S7Z0	TRAF2 and NCK interacting kinase	4.0-6.0 kDa	Actin family	configured audit family match: Actin family
4.96	GIP	P09680	Gastric inhibitory polypeptide	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01	ASNSD1	E1BB20	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.01		A0ABI0NQN9	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.02	TMSB4	A0A3Q1LXR1	Thymosin beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.05	TMSB4	P62326	Thymosin beta-4	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.07	IGIP	P0C712	IgA-inducing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.08		P01154	Posterior pituitary peptide	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.10		A0ACM8QI36	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.17		A0A3Q1MIL4	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks

MW kDa	Gene	Accession	Protein	MW bin	Families	Audit note
5.18		A0ABI0P1U5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.20		A0A3Q1LTD7	Thymosin beta	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.22		A0ABIONM07	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.25	MRLN	A0A3Q1MCJ9	Myoregulin	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.28		A0ABIONT44	Pyrin domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.28	STMP1	A0A3Q1MC14	Short transmembrane mitochondrial protein 1	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.31		A0ABI0P1H3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.34	CHD9NB	A0AAA9TKE9	CHD9 neighbor	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.37		A0AAA9T4M4	Immunoglobulin V-set domain-containing protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.41		A0ABIONUI5	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.44	KRTAP19-2	A0AAA9SLU8	Keratin associated protein 19-2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.46		A0AAA9SS72	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.50		A0ABI0P702	Plakophilin 4	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.57		P59760	Glycolactin	4.0-6.0 kDa	Actin family	configured audit family match: Actin family
5.57		P01250	Thymopoietin-2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.58		P01249	Thymopoietin-1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.64	SP	P01251	Splenin	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.65		O18815	Beta-defensin C7 (Fragment)	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.70		A0AAA9SSC2	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.78	ATP5F1E	P05632	ATP synthase F(1) complex subunit epsilon, mitochondrial	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.86	LOC132345003	A0A3Q1LN99	ATP synthase subunit epsilon, mitochondrial	4.0-6.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
5.86		A0AAA9SMZ3	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.88		A0ABIONL66	Metallothionein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.90	CTXND2	A0AAA9TCX7	Cortexin domain containing 2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.95	MT1A	P67983	Metallothionein-1A	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.96		P55943	Metallothionein-II, hippocampal	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.96	LOC101907769	A0A3Q1MSQ3	Metallothionein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.97		A0A452DJ93	Uncharacterized protein	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.98	ITIH2	P56651	Inter-alpha-trypsin inhibitor heavy chain H2	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.99	MT1	P58280	Metallothionein-1	4.0-6.0 kDa		standard full-blot and antibody-validation checks
5.99	PRCD	E1B7R9	Photoreceptor disk component PRCD	4.0-6.0 kDa		standard full-blot and antibody-validation checks
6.03	MT2A	P68301	Metallothionein-2	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.04	TOMM5	A8YXZ8	Mitochondrial import receptor subunit TOM5 homolog	6.0-8.0 kDa	Mitochondrial markers	configured audit family match: Mitochondrial markers
6.05	CYFIP1	P85091	Cytoplasmic FMR1-interacting protein 1	6.0-8.0 kDa	Actin family	configured audit family match: Actin family
6.05	DEFB9	P46167	Beta-defensin 9 (Fragment)	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.05	PLN	A4IFH6	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.06	LOC613358	A0A3Q1M6T1	Metallothionein	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.07	RELN	Q9N117	Reelin (Fragment)	6.0-8.0 kDa	NF-kB/Rel	configured signaling gene; check phospho/total controls and antibody validation; configured audit family match: NF-kB/Rel
6.07	PMAIP1	A0A3Q1NFP6	Phorbol-12-myristate-13-acetate-induced protein 1	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.08	PLN	F22414	Phospholamban	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09		P55942	Metallothionein-I, hippocampal	6.0-8.0 kDa		standard full-blot and antibody-validation checks
6.09	ZG16B	G3MZ19	Uncharacterized protein	6.0-8.0 kDa		standard full-blot and antibody-validation checks

Reported western-blot malpractice cases mapped to MW bins

These are reported public cases supplied in the malpractice input TSV. They are evidence examples for integrity-audit results sections, not substitution recommendations.

Case	Claim/protein	MW bin	Reported issue	Results sentence
ORI_Eckert_JBC2014_MEK3	MEK3 western blot	38.0-40.0 kDa	ORI reported MEK3 bands were compiled from a source image representing p44 expression in an unrelated experiment.	A public ORI case documents a human-cell western blot malpractice example in the 38-40 kDa bin, where MEK3-labeled bands were compiled from an unrelated p44 source image.
ORI_Eckert_JBC2014_p38d	p38δ western blot	40.0-42.0 kDa	ORI reported p38δ bands were compiled from a source image representing β-actin expression in an unrelated experiment.	The 40-42 kDa bin is supported by a public ORI case in which p38δ-labeled bands were compiled from a β-actin source image.
ORI_Eckert_PLoS2012_TAM67	TAM67-FLAG western blot	48.0-50.0 kDa	ORI reported TAM67-FLAG bands were generated using unrelated bands from a source image representing Cyclin A expression.	A public ORI case maps a relabeled western blot problem to the 48-50 kDa neighborhood, involving Cyclin A source material used for a TAM67-FLAG claim.
ORI_Yang_GAPDH_NFYA	GAPDH loading-control bands	36.0-38.0 kDa	ORI reported claimed GAPDH loading-control bands were actually a prominent nonspecific/background band from an NF-YA western blot.	The 36-38 kDa neighborhood is supported by an ORI-described case where a nonspecific NF-YA blot band was claimed as GAPDH.
CancerGeneTherapy_Liu2021_CTBP2_GAPDH	CTBP2 and GAPDH western blots	48.0-50.0 kDa	Retraction note reported apparent similarity in western blots presented for different proteins, including CTBP2 and GAPDH.	A retraction notice maps an apparent different-protein western blot similarity to the CTBP2/GAPDH region, spanning roughly the 36-38 and 48-50 kDa neighborhoods.
CancerGeneTherapy_Liu2021_CyclinD1_CDK4	Cyclin D1 and CDK4 western blots	34.0-36.0 kDa	Retraction note reported apparent similarity in western blots presented for Cyclin D1 and CDK4.	A retraction notice maps an apparent different-protein western blot similarity to the 32-36 kDa cell-cycle neighborhood involving Cyclin D1 and CDK4.
Das_UConn_Garlic_AKT	Akt / p-Akt western blots in Figure 6	54.0-56.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 54-56 kDa signaling region is supported as an audit-priority bin by the garlic-paper retraction, where Akt/p-Akt figure panels were within figures reported to contain fabricated data.
Das_UConn_Garlic_FOXO1	FoxO1 / p-FoxO1 western blots in Figure 6	70.0-72.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 70 kDa transcription-factor signaling region can be referenced as an audit-priority bin because FoxO1 panels occurred in a figure reported to contain fabricated data.
Das_UConn_Garlic_NFKB_PPAR_Glut4	Nrf2, NF-κB p65, PPARα/δ, Glut-4 western blots in Figures 7 and 8	52.0-54.0 kDa	UConn/ACS retraction materials reported data fabrication in Figures 6, 7, and 8; exact lane-level source images are not public.	The 50-56 kDa metabolic/signaling region is an audit-priority region in the garlic-paper case because PPAR and Glut-4 panels were in figures reported to contain fabricated data.
Ilangovan_HSP25_HSP27	Hsp25/Hsp27-family western blots	22.0-24.0 kDa	Public investigation materials described Hsp-family western blot relabeling and unreliable/falsified blot images.	The 22-24 kDa small heat-shock-protein neighborhood can be cited as an audit-risk bin based on public investigation materials describing Hsp-family western blot relabeling.

Integrity-audit checklist

- Ask for the full, uncropped blot with ladder and lane labels. Cropped panels are not enough.
- Confirm expected molecular weight and apparent molecular weight. Do not treat molecular-weight agreement as proof of identity.
- For configured marker proteins, request source-imager metadata and original antibody-validation records.
- Check whether a marker band is near common markers such as GAPDH, actin, tubulin, HSPs, lamins, VDAC, PCNA, or other configured controls.
- For signaling proteins, verify phospho/total controls, controls for activation state, and an independent identity control when central to the claim.
- In crowded or high-audit bins, prioritize raw-data provenance, full blot context, and orthogonal validation.
- Interpret mapped malpractice cases only as documented audit precedents. They do not imply that proteins in the same bin are interchangeable.

Generated files

- proteome_wb_integrity_atlas.csv - complete per-protein atlas sorted by theoretical MW.
- mw_bin_summary.csv - complete molecular-weight-bin audit summary.
- reported_malpractice_bins.csv - generated when a malpractice TSV is supplied.
- plots/ - four standalone PNG figures embedded in this report.
- western_blot_integrity_atlas_full_report_with_plots.pdf - this consolidated report.
- western_blot_integrity_atlas_report.pdf - byte-identical compatibility copy of the consolidated report.

Input configuration

Input	File
Proteome FASTA	cow_UP000009136_9913.fasta
Marker file	cow_marker_genes.tsv
Signaling-prefix file	cow_signaling_prefixes.tsv
Family-pattern file	cow_family_patterns.tsv
Species/report label	Bos taurus